



Monetary Authority of Singapore

Consultation Paper
P009-2026 – April 2026

Consultation Paper on the Prudential Treatment of Cryptoassets on Permissionless Blockchains



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1. Preface

- 1.1 The Monetary Authority of Singapore (MAS) seeks feedback on a more risk-sensitive prudential treatment for cryptoassets issued on permissionless blockchains.
- 1.2 This consultation paper follows from feedback received on our consultation paper on the prudential treatment of cryptoasset exposures and requirements for additional Tier 1 and Tier 2 capital instruments for banks issued on 27 March 2025¹ (“March 2025 consultation”) that the proposed prudential treatment of cryptoassets on permissionless blockchains (“permissionless cryptoassets”) was not technology neutral and punitive, in view of advances in implementation practices, including technological safeguards through layer 2 solutions which could potentially mitigate the risks arising from the use of permissionless blockchains.
- 1.3 MAS proposes to allow banks to classify and treat a permissionless cryptoasset as a Group 1 cryptoasset, if the permissionless cryptoasset is able to meet a set of principle-based requirements designed to adequately mitigate the risks arising from the use of permissionless blockchains.
- 1.4 The proposed Group 1 prudential treatment for permissionless cryptoassets is intended to be made available to banks from the time of publication of this consultation paper, until otherwise subsequently modified arising from the feedback, or the finalisation of the cryptoasset prudential framework and its effective implementation date in Singapore, whichever is the sooner (“interim period”). During this interim period, the application of this Group 1 prudential treatment will be subject to prescribed exposure and issuance caps.
- 1.5 MAS invites comments from banks and other interested parties. Please refer to Annex A for the list of questions.
- 1.6 Please note that all submissions received will be published and attributed to the respective respondent unless they expressly request MAS not to do so. As such, if respondents would like:
 - (a) their whole submission or part of it (but not their identity), or
 - (b) their identity along with their whole submission,to be kept confidential, please expressly state so in the submission to MAS. MAS will only publish non-anonymous submissions. In addition, MAS reserves the right not to publish any submission

¹ The consultation paper on the prudential treatment of cryptoasset exposures and requirements for additional Tier 1 and Tier 2 capital instruments for banks can be found at <https://www.mas.gov.sg/-/media/mas-media-library/publications/consultations/ppd/2025/consultation-on-the-prudential-treatment-of-cryptoassets-and-reqs-for-at1-and-t2-capital-instruments.pdf>.



received where MAS considers it not in the public interest to do so, such as where the submission appears to be libelous or offensive.

- 1.7 Please submit written comments by 18 May 2026 via the link provided: [FormSG Link](#).
- 1.8 Should you encounter any technical difficulties in your submission, please send your enquiry to prudential_policy_dept@mas.gov.sg.



2. Prudential Treatment of Permissionless Cryptoassets

2.1 Under the proposed prudential treatment in our March 2025 consultation, in order for a cryptoasset to qualify as a Group 1 cryptoasset, all four classification conditions as set out in paragraphs 9A.2.10 to 9A.2.23 of the draft rules must be met.² This meant that permissionless cryptoassets would not qualify as Group 1 cryptoassets as they would not be able to meet specific portions of the classification conditions. These are –

(a) in relation to classification condition 3 –

- (i) the requirement that the network of the cryptoasset, including the distributed ledger technology or similar technology on which the cryptoasset is based, does not pose any material risk that could impact the transferability, settlement finality or, where applicable, redeemability of the cryptoasset and that the entities performing activities associated with the functions of the cryptoasset, specifically the node validators, follow robust risk governance and risk control policies and practices to address risks as specified in paragraph 9A.2.21(a) of the draft rules; and
- (ii) the requirement that all key elements of the network on which the cryptoasset operates are well-defined such that all transactions and participants are traceable as specified in paragraph 9A.2.21(b) of the draft rules;

(b) in relation to classification condition 4 –

- (i) the requirement that the node validators must be regulated and supervised, or subject to appropriate risk management standards as specified in paragraph 9A.2.22(a) of the draft rules; and
- (ii) the requirement that the node validators must have in place a comprehensive governance framework, which must include an accountability and reporting structure and risk control policies and practices for the activities associated with the functions of the cryptoasset, and the governance framework must be publicly disclosed as specified in paragraph 9A.2.22(c) of the draft rules.

2.2 Banks would therefore have to classify all permissionless cryptoassets as Group 2 cryptoassets and subject their exposures to permissionless cryptoassets to the conservative Group 2 prudential treatment. While respondents to the March 2025 consultation acknowledged that cryptoassets

² An extract of the relevant rules is provided in Annex B for easy reference.



operating on permissionless blockchains could introduce novel risks compared to traditional assets, they argued that permissionless cryptoassets should not be precluded from being treated as Group 1 cryptoassets if it could be demonstrated that the relevant risks have been adequately mitigated by appropriate safeguards.

2.3 MAS has carefully considered the feedback and agrees with respondents that permissionless cryptoassets should not be ruled out from being treated as Group 1 cryptoassets if the relevant risks have been addressed. In particular, a bank may be exposed to the following key risks when the bank takes a position in or issues a permissionless cryptoasset³:

- (a) **Governance risk:** Operational integrity of the blockchain may be compromised if the governance authority is concentrated in entities that operate a significant number of the nodes of the permissionless blockchain. This is exacerbated if the governance mechanisms of the blockchain are not sufficiently robust and not transparent to users because this means there is a lack of clear accountability.
- (b) **Technology risk:** Permissionless blockchains are vulnerable to attacks that are unique to the infrastructure of a permissionless blockchain. Examples include the risk of 51% attacks.
- (c) **Settlement finality risk:** Settlement finality depends on the consensus mechanism of each blockchain. For some permissionless blockchains, settlement is probabilistic, which undermines settlement finality as posted transactions could be reversed and invalidated later.
- (d) **Anti-Money Laundering/Countering the Financing of Terrorism (“AML/CFT”) risk:** A bank may transact with malicious or sanctioned entities because permissionless blockchains allow for the pseudonymisation of participants. AML/CFT risk can translate to prudential risks for a bank. Specifically, permissionless cryptoassets are pseudo-anonymous in nature and are more susceptible to being used to facilitate money laundering and illicit finance. Prudential risk could stem from the potential reputational and operational risks that the bank would face if it transacted in a permissionless cryptoasset that poses AML/CFT risks.

2.4 The rules in paragraphs 2.1(a) and (b) were intended to aid in addressing these risks. However, MAS appreciates that these are not the only means through which such risks can be addressed. MAS therefore proposes to disapply the rules set out in those paragraphs (i.e. Paragraphs 9A.2.21(a) and (b) in respect of the network the cryptoasset operates on and the node validators, 9A.2.22(a) and (c) in respect of node validators) for permissionless cryptoassets, provided that the principle-based measures as set out in Annex C to mitigate the governance, technology, settlement finality and AML/CFT risks are

³ Please refer to BCBS working paper 44 titled “Novel risks, mitigants and uncertainties with permissionless distributed ledger technologies” published in August 2024.



met. We intend for this position to continue to apply until the rules for the prudential treatment of cryptoassets are finalised.

- 2.5 To help banks obtain more certainty in their assessments of whether a cryptoasset meets the principle-based requirements in Annex C, we have specified a list of deeming provisions for each principle-based requirement in Annex D. While the cryptoasset ecosystem, its risks and safeguards are evolving rapidly, these are examples of detailed safeguards that may address the risk outcomes of the corresponding principle-based requirement. Where the deeming provisions are met, the cryptoasset is deemed to have met the corresponding principle-based requirement.
- 2.6 If the deeming provisions are not met, a bank must demonstrate, to MAS' satisfaction, that the permissionless cryptoasset meets the principle-based requirements specified in Annex C via other forms of safeguards, before the permissionless cryptoasset can be classified as a Group 1 cryptoasset.

Question 1. *We have proposed to disapply specific portions of the classification conditions which would disqualify all permissionless cryptoassets from being classified as Group 1 and set out proposed principle-based requirements and deeming provisions specified in Annexes C and D respectively to allow permissionless cryptoassets that have adequately mitigated the ensuing risks to be classified as Group 1. MAS seeks views on such an approach and whether there are other aspects of the classification conditions that need to be adjusted to provide banks the leeway to hold exposures to permissionless cryptoassets or engage in issuances of permissionless cryptoassets, while managing the risks.*

Question 2. *MAS seeks views on the proposed principle-based requirements and deeming provisions that are set out in Annexes C and D respectively, whether they are sufficient and appropriate to mitigate the risks arising from permissionless blockchains. If respondents are of the view that there are additional requirements or deeming provisions that should be included, or conversely that there are requirements or deeming provisions which can be modified to manage the risks arising from permissionless blockchains in a different way, please provide the relevant details.*

Question 3. *MAS seeks views on the inclusion of principle-based requirements for AML/CFT risk.*

Exposure and Issuance Caps

- 2.7 To ensure that the risks arising from permissionless cryptoassets that are classified as Group 1 cryptoassets are contained during the interim period till the rules for cryptoassets are finalised, exposure and issuance caps for both locally incorporated banks and bank branches in Singapore⁴ will apply.

⁴ The application of the exposure and issuance caps would be relevant for bank branches in Singapore given that the classification of cryptoassets would impact the calculation of Minimum Liquid Assets (MLA) and Liquidity Coverage Ratio (LCR) requirements, which the bank branches in Singapore are subject to.



2.8 The proposed exposure and issuance caps for a locally-incorporated bank are as follows –

Exposure cap

- (a) A bank must subject its exposures to permissionless cryptoassets that are classified as Group 1 cryptoassets to an exposure cap equivalent to 2% of the bank's Tier 1 capital.

Issuance cap for liabilities

- (b) A bank must subject its issuances of permissionless cryptoassets that result in liabilities for the bank and are classified as Group 1 cryptoassets to an issuance cap equivalent to 5% of the bank's Tier 1 capital.

2.9 The proposed exposure and issuance caps for a bank branch in Singapore ("Singapore branch") are as follows –

Exposure cap

- (a) A bank must subject its exposures to permissionless cryptoassets booked in the Singapore branch that are classified as Group 1 cryptoassets to an exposure cap equivalent to 0.2% of the total assets in the Singapore branch.

Issuance cap for liabilities

- (b) A bank must subject its issuances of permissionless cryptoassets booked in the Singapore branch that result in liabilities for the bank and are classified as Group 1 cryptoassets to an issuance cap equivalent to 1% of the total assets in the Singapore branch.

2.10 For the purposes of determining the cryptoasset exposures that are to be assessed against the exposure caps in paragraphs 2.8(a) and 2.9(a) –

- (a) a bank must measure each cryptoasset exposure as follows:
- (i) for a direct or indirect holding of a cryptoasset, the market value of the cryptoasset; and
 - (ii) for a derivative referencing a cryptoasset, the delta-weighted position calculated as follows:

$\text{Delta-weighted position} = \text{Market value of the underlying cryptoasset} \times \text{delta}^5$
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- (b) the bank may net long and short exposures that are measured in accordance with sub-paragraph (a), if the exposures are to the same cryptoasset; and
- (c) the bank must not consider issuances of cryptoassets as short exposures for the purposes of calculating the exposures in sub-paragraph (a).

⁵ This is the linear estimate of the change in the value of an instrument due to a change in value of the underlying risk factor, which in this case would be the price of the underlying permissionless cryptoasset.



- 2.11 To avoid doubt, the proposed exposure and issuance caps are not intended to limit banks' holdings and issuances of permissionless cryptoassets but are instead intended to limit the amount of permissionless cryptoassets that are eligible to be treated as Group 1 cryptoassets. In the event that a bank has exposures to permissionless cryptoassets or issuances of permissionless cryptoassets that are in excess of the exposure or issuance caps respectively, the bank must classify the excess exposures or issuances, as the case may be, as Group 2 cryptoassets and subject them to the Group 2 prudential treatment, even if the permissionless cryptoasset is able to meet the principle-based requirements and is eligible to be classified as a Group 1 cryptoasset.
- 2.12 MAS will review further the need to apply exposure and issuance caps and if relevant, their calibration, when the rules are finalised and implemented.

Question 4. MAS seeks views on the necessity for exposure and issuance caps to apply when the rules are finalised and implemented. Should respondents be of the view that there is a need for such caps when the rules are finalised and implemented, MAS also seeks views on the calibration of the exposure and issuances caps. Please justify with elaborations where relevant.

Proposed interim treatment till the cryptoasset rules are implemented

- 2.13 During the interim period, a bank is allowed to classify and treat a permissionless cryptoasset as a Group 1 cryptoasset if the following conditions are met:
- (a) The bank must notify MAS at least one month prior to the date the bank intends to classify and treat the permissionless cryptoasset as a Group 1 cryptoasset. When notifying MAS, the bank must also provide a written confirmation from the bank's executive officer responsible for risk management that the permissionless cryptoasset, along with its underlying blockchain, which the bank intends to classify under as a Group 1 cryptoasset, meets all the principle-based requirements specified in Annex C and the deeming provisions specified in Annex D (if applicable);
 - (b) Despite sub-paragraph (a), in cases where the deeming provisions are not met, the bank must demonstrate to MAS' satisfaction that the permissionless cryptoasset meets the principle-based requirements specified in Annex C via other forms of safeguards and seek MAS' prior approval for the classification;
 - (c) The bank must ensure that –
 - (i) its exposures to permissionless cryptoassets that apply this treatment do not exceed the exposure cap specified in paragraph 2.8(a) in the case of locally-incorporated banks or the exposure cap specified in paragraph 2.9(a) in the case of bank branches in Singapore; and



- (ii) its issuances of permissionless cryptoassets that apply this treatment do not exceed the issuance cap specified in paragraph 2.8(b) in the case of locally-incorporated banks or the issuance cap specified in paragraph 2.9(b) in the case of bank branches in Singapore.

2.14 In the case where a bank has existing exposures or issuances of a permissionless cryptoasset that is classified as a Group 2 cryptoasset, the bank is allowed to reclassify and treat the permissionless cryptoasset as a Group 1 cryptoasset if the bank ensures that the conditions specified in paragraph 2.13 are met.

2.15 To avoid doubt, a bank is only required to submit a single notification for each permissionless cryptoasset. The bank must submit a new notification to MAS if there are any changes to the design of the permissionless cryptoasset⁶ or its underlying blockchain.

2.16 MAS retains the right to override the bank's classification of a cryptoasset at any time if the Authority disagrees with the bank's classification.

⁶ For example, changes in the safeguards applied by the cryptoasset.



Annex A: List of Questions

- Question 1.** We have proposed to disapply specific portions of the classification conditions which would disqualify all permissionless cryptoassets from being classified as Group 1 and set out proposed principle-based requirements and deeming provisions specified in Annexes C and D respectively to allow permissionless cryptoassets that have adequately mitigated the ensuing risks to be classified as Group 1. MAS seeks views on such an approach and whether there are other aspects of the classification conditions that need to be adjusted to provide banks the leeway to hold exposures to permissionless cryptoassets or engage in issuances of permissionless cryptoassets, while managing the risks. 7
- Question 2.** MAS seeks views on the proposed principle-based requirements and deeming provisions that are set out in Annexes C and D respectively, whether they are sufficient and appropriate to mitigate the risks arising from permissionless blockchains. If respondents are of the view that there are additional requirements or deeming provisions that should be included, or conversely that there are requirements or deeming provisions which can be modified to manage the risks arising from permissionless blockchains in a different way, please provide the relevant details. 7
- Question 3.** MAS seeks views on the inclusion of principle-based requirements for AML/CFT risk. 7
- Question 4.** MAS seeks views on the necessity for exposure and issuance caps to apply when the rules are finalised and implemented. Should respondents be of the view that there is a need for such caps when the rules are finalised and implemented, MAS also seeks views on the calibration of the exposure and issuances caps. Please justify with elaborations where relevant. 9



Annex B

Classification Conditions



PART IXA: CRYPTOASSETS

This Annex lays out paragraphs 9A.2.10 to 9A.2.23 of Part IXA as provided in the March 2025 Consultation. There are no changes to the draft rules.

Sub-division 2: Classification Conditions

Classification condition 1

9A.2.10 For a cryptoasset to meet classification condition 1, a Reporting Bank must ensure that the cryptoasset either –

- (a) is a tokenised traditional asset that meets all of the following criteria:
 - (i) the cryptoasset is a digital representation of a traditional asset using –
 - (A) cryptography; and
 - (B) DLT or similar technologies, to record ownership; and
 - (ii) subject to paragraph 9A.2.11, the cryptoasset poses the same level of credit and market risk as the traditional asset, as demonstrated by the following:
 - (A) in the case where the cryptoasset is a digital representation of a bond, loan, claim on a bank (including in the form of a deposit)^{928C}, equity or derivative –
 - (I) ownership of the cryptoasset confers the same legal rights as ownership of the traditional asset; and
 - (II) there is no feature of the cryptoasset that would prevent obligations arising from the cryptoasset from being paid in full when due as compared to the traditional asset;
 - (B) in the case where the cryptoasset is a digital representation of a commodity, the cryptoasset confers the same legal rights as an account-based record of ownership of the commodity;

^{928C} To avoid doubt, a tokenised asset that is issued by a bank and is, or is intended to be, a medium of exchange accepted by the public, or a section of the public, as a means of payment, and is backed by the assets on the balance sheet of the bank that are not reserve assets, may represent a claim on the bank and qualify as a tokenised traditional asset, subject to the tokenised asset meeting the criteria set out in paragraph 9A.2.10(a), and regardless of how it is referred to by any country or jurisdiction.



- (C) in the case where the cryptoasset is a digital representation of cash, the cryptoasset confers the same legal rights as cash; or
- (b) has a stabilisation mechanism that is effective at all times in linking its value to a traditional asset or a pool of traditional assets, as demonstrated by meeting all of the following requirements:
 - (i) the cryptoasset is designed to be redeemable for a predefined amount of a reference asset or reference assets^{928D}, or cash equal to the current market value of the reference asset or reference assets^{928E};
 - (ii) the stabilisation mechanism is designed to minimise fluctuations in the market value of the cryptoasset relative to the peg value;
 - (iii) the Reporting Bank must have a framework in place to monitor the effectiveness of the stabilisation mechanism and verify that the stabilisation mechanism is functioning as intended in accordance with the framework;
 - (iv) the Reporting Bank must, based on sufficient data, assess that the stabilisation mechanism functions in a manner that allows the Reporting Bank to conduct risk management activities for the cryptoasset similarly to the risk management of traditional assets;
 - (v) the Reporting Bank must ensure that there is sufficient information available to verify the ownership rights of the reserve assets upon which the stable value of the cryptoasset is dependent, to ensure the reserve assets are unencumbered and can be used to back the value of the cryptoasset. Where the reserve assets are physical assets, the Reporting Bank must additionally verify that the physical assets are stored and managed appropriately;
 - (vi) the cryptoasset passes the redemption risk test set out in paragraph 9A.2.14;
 - (vii) the issuer of the cryptoasset is licensed, approved, registered or otherwise regulated by the Authority, or by a financial regulatory agency of a country or jurisdiction other than Singapore, that applies prudential capital and liquidity requirements to the issuer.

9A.2.11 A Reporting Bank must not consider a cryptoasset to meet the criterion set out in paragraph 9A.2.10(a)(ii) if –

- (a) the cryptoasset first needs to be redeemed for or converted into a traditional asset before it receives the same legal rights as direct ownership of the traditional asset; or

^{928D} For example, 1 USD or 1 oz gold.

^{928E} For example, the USD value of 1 oz gold.

- (b) the cryptoasset's structure or arrangement exposes the holder to additional counterparty credit risk relative to the traditional asset.

9A.2.12 Where there is less than 12 months between the time of first issuance of a cryptoasset and the time of screening against the classification conditions, in order for the cryptoasset to be considered as meeting the requirement set out in paragraph 9A.2.10(b)(iv), the Reporting Bank must document the assessment conducted and evidence used, to determine the effectiveness of the stabilisation mechanism and provide the assessment and evidence to the Authority, upon request. The Reporting Bank must ensure that the evidence minimally includes information on the composition, valuation and frequency of valuation of the reserve assets and the quality of available data.

9A.2.13 For the purposes of paragraph 9A.2.10(b)(v), a Reporting Bank may rely on the assessments of independent third parties^{928F}, provided that the Reporting Bank ensures that the assessments are reliable and the Reporting Bank documents the basis for its satisfaction that the third party's assessment is reliable.

9A.2.14 To pass the redemption risk test^{928G} referred to in paragraph 9A.2.10(b)(vi), a Reporting Bank must satisfy itself that a cryptoasset meets all of the following conditions:

- (a) where the cryptoasset references one or more currencies –
 - (i) the reserve assets of the cryptoasset comprise only the following:
 - (A) reserves held by the issuer with MAS or other central banks, to the extent that MAS' or the central banks' policies, as the case may be, allow them to be drawn down in times of stress;
 - (B) any marketable security representing a claim on, or that is guaranteed by a central government, a central bank, a PSE, the Bank for International Settlements, the International Monetary Fund, the European Central Bank, the European Union, the European Stability Mechanism, the European Financial Stability Facility, or an MDB, which satisfies the following conditions:
 - (I) it is assigned a 0% risk-weight under Table 7-1 or paragraphs 7.3.46 to 7.3.51;
 - (II) it is traded in large, deep and active repo or cash markets characterised by a low level of concentration;
 - (III) it has a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions;

^{928F} For example, the depository or the custodian bank.

^{928G} The objective of the redemption risk test is to ensure that the reserve assets are sufficient to enable the cryptoasset to be redeemable at all times for the peg value, including during periods of extreme market stress.



- (IV) it is not an obligation of a financial institution or any of its related corporations;
- (C) where a central government or central bank is assigned a 20% risk weight under Table 7-1, any marketable security representing a claim on, or that is guaranteed by the central government or central bank and that is denominated in the domestic currency of the country or jurisdiction of that central government or central bank and which satisfies the following conditions:
 - (I) it is traded in large, deep and active repo or cash markets characterised by a low level of concentration;
 - (II) it has a proven record as a reliable source of liquidity in the markets (repo or sale) even during stressed market conditions,to the extent to which the domestic currency of that central government or central bank is referenced by the cryptoasset^{928H};
- (D) any cash receivables from reverse repos that have residual maturities of one week or less and that are overcollateralised by the securities specified in sub-paragraph (a)(i)(B) or (a)(i)(C), or both. To avoid doubt, the collateral underlying the reverse repos must not be included as reserve assets;
- (E) cash deposits at banks that have a credit quality grade of “2” or better as set out in Table 7M-1 by a recognised ECAI and that meet at least one of the following conditions:
 - (I) the bank is regulated by a bank regulatory agency that subjects the bank to the standards under the Basel framework promulgated by the BCBS, including the liquidity coverage ratio;
 - (II) the bank applies, on a consolidated basis, concentration limits on its sources of funding;
- (ii) the reserve assets have a proven record of relative stability of value^{928I} even during stressed market conditions;

^{928H} To avoid doubt, the proportion of reserve assets that can be met with marketable securities that represent claims on or are guaranteed by the central government or central bank and that are denominated in the domestic currency of the central government or central bank, is limited to the weight that the domestic currency represents in the peg value.

^{928I} For example, the reserve assets have a proven record of low volatility of traded prices or spreads.

- (iii) the portfolio weighted average residual maturity of the pool of reserve assets upon which the stable value of the cryptoasset is dependent does not exceed 3 months;
- (iv) the reserve assets have sufficient daily liquidity to meet instant redemption requests from the cryptoasset holders;
- (v) the reserve assets are held in structures such that they are bankruptcy remote from any party that issues the cryptoasset, manages or performs a function in the cryptoasset arrangement or custodies the reserve assets;
- (vi) despite sub-paragraph (a)(v), in the case of cash deposits that are specified in sub-paragraph (a)(i)(E), the cash deposits are not required to be bankruptcy remote from the bank where the cash deposits are held if the bank is only involved in the provision of custody services under the cryptoasset arrangement;
- (vii) the value of the reserve assets (net of all claims on the reserve assets that are not associated with the redemption of the cryptoasset) sufficiently overcollateralises the redemption rights of all issuances of the cryptoasset in circulation (referred to in this paragraph as “outstanding cryptoassets”). The level of overcollateralisation must be sufficient to ensure that the value of the reserve assets (net of all claims on the reserve assets that are not associated with the redemption of the cryptoasset) equals or exceeds the aggregate peg value of all outstanding cryptoassets at all times, including during periods of extreme market stress and periods of volatile markets, and account for risks to the cryptoasset holders that are additional to the risks arising from the reference asset or reference assets^{928J};
- (viii) the reserve assets are denominated in –
 - (A) the same currency as the reference asset of the cryptoasset, where the cryptoasset has a currency as its reference asset; or
 - (B) the currencies in the same ratio as the currencies used as the reference assets of the cryptoasset, where the cryptoasset has more than one currency as its reference asset;
- (ix) despite sub-paragraph (a)(viii), a de minimis portion of the reserve assets may be held in a currency other than the currencies referred to in sub-paragraph (a)(viii)(A) or (a)(viii)(B), as the case may be, provided that the holding of such currency is necessary for the

^{928J} For example, for a cryptoasset that is redeemable for a given currency amount but which is backed by marketable securities denominated in the same currency as the reference asset, the reserve assets will give rise to credit, market and liquidity risks of the marketable security that may result in losses relative to the value of the reference asset.

operation of the cryptoasset arrangement and all currency mismatch risk between the reserve assets and the peg value has been appropriately hedged. In the case where the currency mismatch risk has been hedged, the collateral specified in the credit support annex agreements must not be included as reserve assets;

- (b) where the cryptoasset does not reference one or more currencies –
 - (i) the reserve assets must consist of only the reference asset(s), with the exception of a de minimis portion of the reserve assets which may be held in cash or bank deposits, provided that the holding of the cash or bank deposits is necessary for the operation of the cryptoasset arrangement;
 - (ii) at all times, including during periods of extreme market stress, the value of the reserve assets (net of all non-cryptoasset claims on the reserve assets) equals or exceeds the aggregate peg value of all outstanding cryptoassets;
- (c) the governance arrangements relating to the management of the reserve assets are comprehensive and transparent, as demonstrated by the following:
 - (i) the reserve assets are managed and invested with the objective of ensuring that all cryptoassets can be redeemed within 5 calendar days of the issuer's receipt of the redemption request and the redemption is made at the peg value at all times, including under periods of extreme market stress;
 - (ii) the cryptoasset arrangement has a robust operational risk and resilience framework in place to ensure the availability and safe custody of the reserve assets to meet redemption obligations;
 - (iii) the cryptoasset arrangement has a mandate that describes the types of assets that may be included as reserve assets and the mandate is publicly disclosed and kept up to date;
 - (iv) the cryptoasset arrangement has a risk management framework to assess and monitor the risks of the reserve assets including, but not limited to, market risk, credit risk, concentration risk and liquidity risk^{928K};
 - (v) the composition of the reserve assets is publicly disclosed at least weekly, and the value of the reserve assets and the outstanding amount of the cryptoasset in circulation is publicly disclosed at least daily. The disclosed information is verified by an independent third party at least semi-annually to ensure its completeness, fairness of valuation and accuracy;

^{928K} Examples include ongoing monitoring of deposit counterparties and custodians, daily valuation of the reserve assets and stress testing.



- (vi) the composition and value of the reserve assets and the outstanding amount of the cryptoasset in circulation are subject to an independent external audit at least annually to confirm they match the disclosures required pursuant to sub-paragraph (c)(v) and are consistent with the mandate.

9A.2.15 A Reporting Bank must perform due diligence to ensure that a cryptoasset meets the condition specified in paragraphs 9A.2.14(a)(vii) or 9A.2.14(b)(ii), as the case may be –

- (a) prior to acquisition of the cryptoasset;
- (b) at least on a semi-annual basis after the acquisition; and
- (c) when the Reporting Bank becomes aware of any development, including information indicating stressed market conditions, that adversely affects or is likely to adversely affect the effectiveness of the stabilisation mechanism of the cryptoasset,

and document the results of the due diligence conducted and provide the results to the Authority prior to the initial classification of the cryptoasset and upon request by the Authority.

9A.2.16 A Reporting Bank must not consider cryptoassets with stabilisation mechanisms that –

- (a) have other cryptoassets (including cryptoassets that have traditional assets as reference assets) as the reference assets; or
- (b) use protocols to increase or decrease the supply of the cryptoassets to maintain their values^{928L},

to meet classification condition 1.

Classification condition 2

9A.2.17 For a cryptoasset to meet classification condition 2, a Reporting Bank must –

- (a) conduct sufficient legal review, including by obtaining a written independent legal opinion^{928M}, to satisfy itself that, at all times, the following requirements are met:

^{928L} Cryptoassets that use protocols to increase or decrease the supply of the cryptoassets to maintain their value may be referred to as “algorithm-based stablecoins”.

^{928M} While a Reporting Bank may use either an in-house or external legal counsel for the purposes of obtaining a written independent legal opinion, it should consider whether or not an in-house counsel opinion is appropriate. In the case of an in-house legal counsel, the Reporting Bank should ensure that the in-house legal counsel is independent of the parties originating transactions giving rise to exposures in the cryptoasset.



- (i) all rights, obligations and interests arising from the cryptoasset arrangement are clearly defined and legally enforceable in all countries or jurisdictions where the cryptoasset is issued and redeemed;
- (ii) the applicable legal frameworks in all relevant countries or jurisdictions support settlement finality in primary markets and in secondary markets;
- (iii) the cryptoasset arrangement provides for full transferability of ownership of the cryptoasset and settlement finality, at all times, and the documentation of the cryptoasset arrangement sets out when the cryptoasset and its financial risks, become irrevocably and unconditionally transferred from one party to another;
- (iv) where the cryptoasset has a stabilisation mechanism, the cryptoasset arrangement –
 - (A) clearly sets out the objective of the management and investment of reserve assets referred to in paragraph 9A.2.14(c)(i);
 - (B) provides a legal claim against the issuer or the reserve assets, or both, and provides for full redeemability at all times at the peg value, where full redeemability means the ability to complete an exchange of a cryptoasset for amounts of pre-defined assets^{928N} within 5 calendar days of the redemption request;
 - (C) clearly defines which parties have the right to redeem, the obligation of the redeemer to fulfil the redemption function as specified under the cryptoasset arrangement, the timeframe for the redemption to take place, the traditional assets in the exchange and how the redemption value is determined; and
 - (D) provides that the legal rights and obligations under sub-paragraphs (a)(iv)(B) and (C) are enforceable even where parties involved in the cryptoasset arrangement are not located in the same country or jurisdiction where the cryptoasset is issued and redeemed; and
- (b) satisfy itself that the documentation on the cryptoasset is publicly disclosed by the issuer of the cryptoasset, or in the case where the cryptoasset is not sold to the public^{928O}, is made available to all potential investors of the cryptoasset, including those that become investors via secondary market sales, the Authority and the bank regulatory agency

^{928N} For example, cash, bonds, commodities, equity or other traditional assets.

^{928O} For example, the cryptoasset is issued via private placement.



other than the Authority, of any bank that is permitted to become an investor.

9A.2.18 The Reporting Bank must ensure that assumptions underlying the legal opinion pursuant to paragraph 9A.2.17(a) are not unduly restrictive, and are specific, of a factual nature and adequately explained within the legal opinion. Where qualifications are made, the Reporting Bank must ensure that such qualifications are specific and their effect is adequately explained within the legal opinion. A Reporting Bank must examine and assess the assumptions and qualifications in the legal opinion.

9A.2.19 If the Reporting Bank determines that any of the assumptions or qualifications in the legal opinion gives rise to reasonable doubt as to the validity or enforceability of rights, obligations and interests arising from the cryptoasset arrangement, the Reporting Bank must not consider the cryptoasset to meet classification condition 2.

9A.2.20 A Reporting Bank must provide to the Authority, upon request, documentation of the legal review conducted and copies of legal opinions obtained pursuant to paragraph 9A.2.17(a).

Classification condition 3

9A.2.21 For a cryptoasset to meet classification condition 3, a Reporting Bank must satisfy itself that the cryptoasset meets the following requirements:

- (a) the functions of the cryptoasset^{928P} and the network on which it operates, including the DLT or similar technology on which it is based, do not pose any material risks that could impact the transferability, settlement finality or, where applicable, redeemability of the cryptoasset. The Reporting Bank must satisfy itself that the entities performing activities associated with the functions of the cryptoasset^{928Q} follow robust risk governance and risk control policies and practices to address risks including, but not limited to –
 - (i) credit, market and liquidity risks;
 - (ii) operational risks including, but not limited to, outsourcing, fraud and cyber risks;
 - (iii) the risk of loss of data; and
 - (iv) non-financial risks including, but not limited to, risks relating to data integrity, operational resilience, reliability and capacity, third-party risk management and AML/CFT;

^{928P} For example, the issuance, validation, redemption and transfer of the cryptoasset.

^{928Q} Examples of these entities include, but are not limited to, issuers, operators of the transfer and settlement systems for the cryptoasset, administrators of the cryptoasset stabilisation mechanism and custodians of any underlying assets supporting the stabilisation mechanism.



- (b) all key elements of the network on which the cryptoasset operates are well-defined such that all transactions and participants are traceable. Key elements of the network on which the cryptoasset operates include –
 - (i) the operational structure of the network (i.e. whether there is one or multiple entities that perform the core function or core functions, as the case may be, of the network);
 - (ii) the degree of access of the network (i.e. whether access to the network is restricted or unrestricted);
 - (iii) the technical roles of the nodes, including whether there is a differential role and responsibility among the nodes; and
 - (iv) the validation and consensus mechanism of the network (i.e. whether the validation of a transaction is conducted with single or multiple entities).

Classification condition 4

9A.2.22 For a cryptoasset to meet classification condition 4, a Reporting Bank must satisfy itself that all entities that execute redemptions or transfers, provide storage, execute settlement of the cryptoasset or manage the reserve assets of the cryptoasset meet all of the following conditions:

- (a) node validators must be regulated and supervised, or subject to appropriate risk management standards;
- (b) all entities falling within the scope of this paragraph, with the exception of node validators, must be regulated and supervised;
- (c) all entities falling within the scope of this paragraph must have in place a comprehensive governance framework, which must include an accountability and reporting structure and risk control policies and practices for the activities associated with the functions of the cryptoasset, and the governance framework must be publicly disclosed.

9A.2.23 To avoid doubt, the entities falling within the scope of paragraph 9A.2.22 include –

- (a) operators of the transfer and settlement systems for the cryptoasset;
- (b) providers of wallets for the storage of the cryptoasset; and
- (c) for cryptoassets with stabilisation mechanisms, administrators of the stabilisation mechanism and custodians of the reserve assets of the cryptoasset.



Annex C

Prudential Treatment of Permissionless Cryptoassets – Principle-based requirements

Requirements for permissionless cryptoassets to be classified as Group 1 cryptoassets

- 1 A bank may classify and treat permissionless cryptoassets as Group 1 cryptoassets if the bank ensures that the permissionless cryptoasset meets all the following requirements at all times:
 - (a) classification conditions 1 to 4 as specified in paragraphs 9A.2.10 to 9A.2.23 of the draft rules in the consultation except –
 - (i) in relation to classification condition 3, the requirement that the network of the cryptoasset, including the distributed ledger technology or similar technology on which the cryptoasset is based does not pose any material risk that could impact the transferability, settlement finality or, where applicable, redeemability of the cryptoasset and the requirement that the node validators must follow robust risk governance and risk control policies and practices to address risks as specified in paragraph 9A.2.21(a) of the draft rules in the consultation and the requirement that all key elements of the network on which the cryptoasset operates are well-defined such that all transactions and participants are traceable as specified in paragraph 9A.2.21(b) of the draft rules in the consultation; and
 - (ii) in relation to classification condition 4, the requirement that node validators must be regulated and supervised, or subject to appropriate risk management standards as specified in paragraph 9A.2.22(a) of the draft rules in the consultation and the requirement that node validators must have in place a comprehensive risk governance framework that is publicly disclosed as specified in paragraph 9A.2.22(c) of the draft rules in the consultation; and
 - (b) the principle-based requirements that are specified in paragraphs 2 to 4.
- 2 To address the governance risk arising from taking a position in or issuing a cryptoasset on a permissionless blockchain, a bank must ensure that all the following requirements are met:
 - (a) The permissionless blockchain has a sufficiently diversified set of validators;
 - (b) The permissionless blockchain operates with comprehensive governance arrangements, which are clearly documented and accessible to users of the blockchain.
- 3 To address the technology risk and settlement finality risk arising from taking a position in or issuing a cryptoasset on a permissionless blockchain, a bank must ensure that all the following requirements are met:



- (a) The cryptoasset issuer has in place systems and processes to ensure the accuracy and integrity of the transaction records;
 - (b) The cryptoasset issuer must have an effective business continuity plan (BCP) in place to ensure the operational resilience of the cryptoasset even if the permissionless blockchain fails.
- 4 To address the AML/CFT risk arising from the use of permissionless blockchains, a bank must ensure that all the following requirements are met:
 - (a) The cryptoasset issuer is regulated for AML/CFT risks and must have in place adequate policies and procedures for customer on-boarding and for the issuance and redemption of the cryptoasset to address AML/CFT risks;
 - (b) The cryptoasset issuer must have in place mechanisms to verify the identity of each cryptoasset holder or in cases where it is not possible to verify the identifies of all cryptoasset holders, appropriate measures that have been demonstrated to be effective in mitigating AML/CFT risks.



Annex D

Prudential Treatment of Permissionless Cryptoassets – Deeming Provisions



Deeming provisions

- 1 A bank is deemed to have fulfilled the requirements in paragraph 2 of Annex C in respect of a cryptoasset on a permissionless blockchain if it ensures that all the following criteria are met:
 - (a) The blockchain must either have a significantly large number of validator nodes or no single entity controls a significant share of the validator nodes or voting power;
 - (b) The blockchain has the following:
 - (i) Monitoring mechanisms to regularly monitor and assess validator node performance and governance mechanisms to detect and penalise malicious behaviour or performance failures of validator nodes;
 - (ii) Dispute resolution mechanisms and consensus mechanisms to address any disagreements between validator nodes to allow for consensus to be reached on decisions relating to the blockchain protocol;
 - (c) The governance arrangements of the blockchain, including the mechanisms specified in subparagraphs (b)(i) and (ii) are documented and made available to users of the blockchain.
- 2 A bank is deemed to have fulfilled the requirement in paragraph 3 of Annex C if it ensures that all the following criteria are met:
 - (a) The cryptoasset issuer maintains control over the cryptoasset through functions, which allow the cryptoasset issuer to correct or freeze transactions on the blockchain where necessary, such as in cases of settlement failure and there are appropriate governance controls in place to ensure that such amendments are not performed unilaterally and that the cryptoasset issuer has the legal rights to perform such amendments;
 - (b) There must be a point of finality defined for the underlying blockchain of the cryptoasset, which is documented and made available to users of the blockchain, where the point of finality refers to a level of confirmation where any reversal of transactions on the blockchain would be economically or technically impractical. All transactions on the blockchain are considered final only when the blockchain specific point of finality has been met or exceeded;
 - (c) The bank must –
 - (i) establish a surveillance process to monitor the health of the underlying blockchain of the cryptoasset and be able to detect on the underlying blockchain, the occurrence of



anomalies, erroneous behaviours and events indicating a potential cyber incident. The bank may rely on third-party vendors to conduct the monitoring, provided that the bank ensures that the information provided by the third-party vendors are reliable and documents the basis for its satisfaction that the information provided by the third party is reliable; and

- (ii) assess the results from the surveillance process on an ongoing basis;
 - (d) Where smart contracts are used, the cryptoasset issuer must engage independent third-party entities such as smart contract auditors to audit the design and code of the smart contract, including any changes to the smart contract to minimise the risk of bugs, vulnerabilities and coding errors;
 - (e) The BCP minimally includes the following:
 - (i) Documented policies on the golden source of transactions in the event of blockchain incidences⁷;
 - (ii) Identification of critical business functions and systems, and well-documented backup and disaster recovery procedures and recovery measures for such business functions and systems;
 - (f) The cryptoasset issuer must –
 - (i) regularly review the BCP to ensure that it remains comprehensive and appropriate; and
 - (ii) regularly test the BCP to ensure it remains fit for purpose;
- 3 A bank is deemed to have fulfilled the requirements in paragraph 4 of Annex C if it ensures that the cryptoasset issuer has the following in place:
- (a) The cryptotasset issuer has put permissioning controls⁸ in place for the cryptoasset, so that only whitelisted entities or whitelisted wallets that have been pre-screened and verified can hold and perform transactions with the cryptoasset. The cryptoasset issuer may rely on third party service providers⁹ to conduct the permissioning controls provided that the cryptoasset issuer has

⁷ An example would be the use of an off-chain database to ensure the availability and recoverability of data in the event of disruptions to the blockchain.

⁸ An example of a permissioning control would be the use of “Trust Anchors”.

⁹ For example, distributors, virtual asset service provider or virtual asset trading platforms.



conducted due diligence on the third party service provider and documents the basis for its reliance on the third party service provider;

- (b) If smart contracts are used to implement the permissioning controls specified in sub-paragraph (a), the cryptoasset issuer must engage independent third-party entities such as smart contract auditors to audit the design and code of the smart contract, including any changes to the smart contract, to minimise the risk of bugs, vulnerabilities and coding errors.